Four equivalent characterizations of an identity system

The Univalent Foundations Program

hott-book:578b85cc8d586b1677ec4335148adeb443057d24; CC BY-SA 3.0 Unported

Editorial note. Complete fixed-author source spans retained; original CC BY-SA attribution and macros reused. Supporting statements do not create extra counted problems.

Editorial note. Construct pointed-predicate contractions and compare their maps with representable identity predicates; recover the dependent elimination rule from contraction of the total space, including its basepoint computation.

Allowed standard prerequisites (editorial).

- Basic univalent type theory. Identity elimination, function extensionality, univalence, dependent sums/products, propositional truncation, standard path transport and the explicitly supplied higher-inductive elimination rules. Standard elementary real/ordinal facts may be used; all target-specific local construction proofs are retained.

Source reference. Original reference 3.11.8: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1760–1760. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Source reference. Original reference 4.7.7: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1891–1891. Original book navigation or standard prerequisite reference; target construction and its local proof chain are retained in body spans.

Definition 5.0.1. Let A be a type and $a_0 : A$ an element.

- A **pointed predicate** over (A, a_0) is a family $R : A \to \mathcal{U}$ equipped with an element $r_0 : R(a_0)$.
- For pointed predicates (R, r_0) and (S, s_0), a family of maps $g : \prod_{(b:A)} R(b) \to S(b)$ is **pointed** if $g(a_0, r_0) = s_0$. We have

$$\mathsf{ppmap}(R, S) :\equiv \sum_{g:\prod_{(b:A)} R(b)\to S(b)} (g(a_0, r_0) = s_0).$$

- An **identity system at** a_0 is a pointed predicate (R, r_0) such that for any type family $D : \prod_{(b:A)} R(b) \to \mathcal{U}$ and $d : D(a_0, r_0)$, there exists a function $f : \prod_{(b:A)} \prod_{(r:R(b))} D(b, r)$ such that $f(a_0, r_0) = d$.

Theorem 5.0.2. *For a pointed predicate (R, r_0) over (A, a_0), the following are logically equivalent.*

(i) (R, r_0) *is an identity system at* a_0.
(ii) For any pointed predicate (S, s_0)*, the type* $\mathsf{ppmap}(R, S)$ *is contractible.*
(iii) For any $b : A$*, the function* $\mathsf{transport}^R(-, r_0) : (a_0 =_A b) \to R(b)$ *is an equivalence.*
(iv) The type $\sum_{(b:A)} R(b)$ *is contractible.*

Proof. First, assume (i) and let (S, s_0) be a pointed predicate. Define $D(b, r) :\equiv S(b)$ and $d :\equiv s_0 : S(a_0) \equiv D(a_0, r_0)$. Since R is an identity system, we have $f : \prod_{(b:A)} R(b) \to S(b)$ with $f(a_0, r_0) = s_0$; hence $\mathsf{ppmap}(R, S)$ is inhabited. Now suppose $(f, f_r), (g, g_r) : \mathsf{ppmap}(R, S)$, and define $D(b, r) :\equiv (f(b, r) = g(b, r))$, and let $d :\equiv f_r \cdot {g_r}^{-1} : f(a_0, r_0) = s_0 = g(a_0, r_0)$. Then again since R is an identity system, we have $h : \prod_{(b:A)} \prod_{(r:R(b))} D(b, r)$ such that $h(a_0, r_0) = f_r \cdot {g_r}^{-1}$. By function extensionality and the characterization of paths in Σ-types and path types, these data yield an equality $(f, f_r) = (g, g_r)$. Hence $\mathsf{ppmap}(R, S)$ is an inhabited mere proposition, and thus contractible; so (ii) holds.

Now suppose (ii), and define $S(b) :\equiv (a_0 = b)$ with $s_0 :\equiv \mathsf{refl}_{a_0} : S(a_0)$. Then (S, s_0) is a pointed predicate, and $\lambda b.\, \lambda p.\, \mathsf{transport}^R(p, r) : \prod_{(b:A)} S(b) \to R(b)$ is a pointed family of maps from S to R. By assumption, $\mathsf{ppmap}(R, S)$ is contractible, hence inhabited, so there also exists a pointed family of maps from R to S. And the composites in either direction are pointed families of maps from R to R and from S to S, respectively, hence equal to identities since $\mathsf{ppmap}(R, R)$ and $\mathsf{ppmap}(S, S)$ are contractible. Thus (iii) holds.

Now supposing (iii), condition (iv) follows from original reference 3.11.8, using the fact that Σ-types respect equivalences (the "if" direction of original reference 4.7.7).

Finally, assume (iv), and let $D : \prod_{(b:A)} R(b) \to \mathcal{U}$ and $d : D(a_0, r_0)$. We can equivalently express D as a family $D' : (\sum_{(b:A)} R(b)) \to \mathcal{U}$. Now since $\sum_{(b:A)} R(b)$ is contractible, we have

$$p : \prod_{u:\sum_{(b:A)} R(b)} (a_0, r_0) = u.$$

Moreover, since the path types of a contractible type are again contractible, we have $p((a_0, r_0)) = \mathsf{refl}_{(a_0,r_0)}$. Define $f(u) :\equiv \mathsf{transport}^{D'}(p(u), d)$, yielding $f : \prod_{(u:\sum_{(b:A)} R(b))} D'(u)$, or equivalently $f : \prod_{(b:A)} \prod_{(r:R(b))} D(b, r)$. Finally, we have

$$f(a_0, r_0) \equiv \mathsf{transport}^{D'}(p((a_0, r_0)), d) = \mathsf{transport}^{D'}(\mathsf{refl}_{(a_0,r_0)}, d) = d.$$

Thus, (i) holds. □